The term *prioritized experience replay* generally refers to a specific implementation introduced in an academic paper titled “Prioritized Experience Replay” by Tom Schaul et al. (2015), and it uses a much more sophisticated mechanism to prioritize experiences. In their implementation, all experiences are recorded just once, unlike our approach, but rather than selecting a mini-batch from the replay completely randomly (i.e., uniformly), they preferentially select experiences that are more informative. They defined informative experiences as not merely those that led to a winning state like we did, but rather those where the DQN had a high error in predicting the reward. In essence, the model preferentially trains on the most surprising experiences. As the model trains, however, the once surprising experiences become less surprising, and the preferences get continually reweighted. This leads to substantially improved training performance. This kind of prioritized experience replay is standard practice for value-based reinforcement learning, whereas policy-based reinforcement learning still tends to rely on using multiple parallelized agents and environments.

11.2.2. Proximal policy optimization (PPO)

We mostly implemented *deep Q-networks* (DQN) in this book rather than policy functions, and this is for good reason. The (deep) policy functions we implemented in chapters 4 and 5 were rather unsophisticated and would not work very well for more complex environments. The problem is not with the policy networks themselves but with the training algorithm. The simple REINFORCE algorithm we used is fairly unstable. When the rewards vary significantly from action to action, the REINFORCE algorithm does not lead to stable results. We need a training algorithm that enforces smoother, more constrained updates to the policy network.

Proximal policy optimization (PPO) is a more advanced training algorithm for policy methods that allows for far more stable training. It was introduced in the paper “Proximal Policy Optimization Algorithms” by John Schulman et al. (2017) at OpenAI. We did not cover PPO in this book because, while the algorithm itself is relatively simple, understanding it requires mathematical machinery that is outside the scope of this introductory book. Making deep Q-learning more stable required only a few